

Context-aware Risk Degree Prediction for Smartphone Zombies

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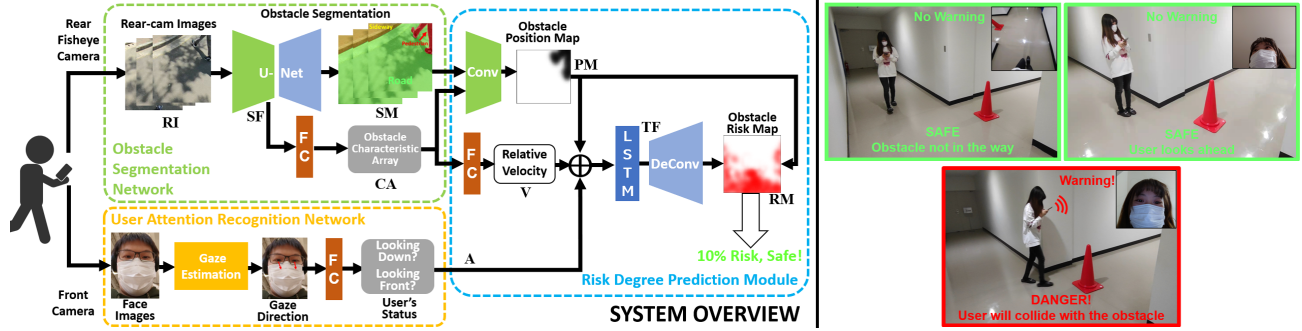


Figure 1: System Overview.

ABSTRACT

Using smartphones while walking is becoming a social problem. Recent works try to support this issue by different warning systems. However, most only focus on detecting obstacles, without considering the risk to the user. In this paper, we propose a deep learning-based context-aware risk prediction system using a built-in camera on smartphones, aiming to notify "smombies" by a risk-degree based algorithm. The proposed system both estimates the risk degree of a potential obstacle and the user's status, which can also be used for distracted driving or visually impaired people.

KEYWORDS

Smombie, Context-aware, Risk Prediction, Deep Learning

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1 INTRODUCTION

Smartphone devices have become indispensable in life, people can use phones anytime anywhere, even while walking. However, smartphone zombies (smombies) are turning into a social problem. Therefore, several warning systems are developed based on activity recognition or object detection, while most just simply notify the users when an obstacle is detected, without considering users' attention or whether the obstacle is really harmful to the user.

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Therefore, in this paper, we propose a deep learning-based context-aware risk degree prediction system using an ultra-wide camera, which is commonly equipped on current smartphones, to support using a smartphone while walking. The proposed system consists of three parts: a segmentation network, a risk prediction network, and an attention recognition network. To train the segmentation network, we build a virtual street simulator with different types of obstacles to collect synthetic image data. The simulator is also used for generating the risk degree maps which are used for the risk prediction. A prototype is developed using the proposed network and a pilot study is conducted to compare with existing methods.

2 RELATED WORK

Most of the existing smartphone technologies for pedestrian safety require extra sensors to recognize the environment, which is not practical. Smombie Guardian [Kim et al. 2018] introduced a method to detect an obstacle using the rear camera, however, they used a simple detection algorithm and could only detect one obstacle at a time. Goh et al. [Goh et al. 2020] are the first who used the word "forecaster", their system predicts potential hazards from the surroundings and users' behavior. However, none of these methods takes the characteristics of an obstacle into account. For example, the risk of a moving vehicle is definitely higher than a stopped one.

On the other hand, real-time prediction from previous information are widely studied. Wu et al. [Wu and Koike 2019] introduced a CNN-based LSTM network to predict future posture from previous motions in real-time, and apply it to martial arts. In terms of first person view cameras, Yagi et al. [Yagi et al. 2018] introduced a three-stream encoder-decoder network to estimate future human locations from past locations. In this paper, we combine the method of Wu et al. and Yagi et al. and developed a multi-stream LSTM network to realize the prediction.

3 IMPLEMENTATION

Fig. 1 shows the overview of the system, which consists of three networks: an obstacle segmentation network, a risk prediction network, and an attention recognition network. The rear camera image

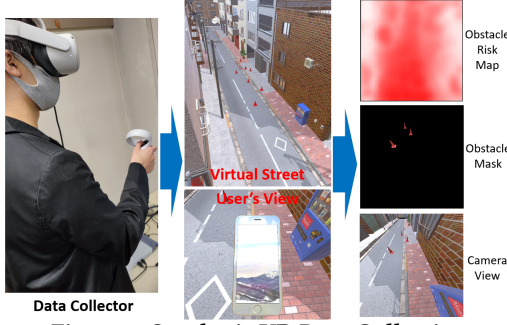


Figure 2: Synthetic VR Data Collection.

sequences **RI** are resized to 256×256 and input to the obstacle segmentation network which consists of a U-Net-based network using ResNet18 as a backbone. Then the convolutional layer extracts the spatial features **SF** from **RI**, while the deconvolutional layers translate **SF** into an obstacle segmentation map **SM** and a fully-connected layer estimates an obstacle characteristic **CA** from **SF**. The output array **CA** includes the obstacle type, distance, and size.

For the attention recognition network, we employ a gaze-estimation module from OpenVINO, the estimated 2D gaze vector is passed to a fully-connected layer to output a boolean value **A** classifying whether the user is looking at the obstacle. Next, to predict the risk degree of the detected obstacle, the mask **SM** and array **CA** are further input to a convolutional layer to obtain the obstacle position map **PM**. Also, a relative velocity value between the obstacle and the user **V** is also estimated by a single fully-connected layer from the array **CA**. The flattened position map **PM** is concatenated with the values **A** and **V** and passed to a LSTM layer to further extract the temporal features **TF**. Finally, a deconvolution layer decodes the **TF** into an obstacle risk map **RM**. The obstacle risk degree can be calculated from the multiplication: $\mathbf{PM} \times \mathbf{RM}$.

4 DATA COLLECTION

For data collection, virtual reality (VR) is employed to let a collector walk freely in the simulation environment (Fig. 2). The subject is holding a VR controller which is linked with a smartphone model so that they can perform smombies in the virtual scene. A camera module is placed on the back of the smartphone of which the parameters are set as the same as the real ultra-wide camera used in our prototype. As a result, the camera images (input of network) and the segmentation mask (GT of the network) can be easily collected. To obtain the risk map (risk matrix), subjects are asked to walk freely in the virtual world while playing a game on the virtual smartphone. The risk map is calculated based on the probability of collision of the direction where the obstacles come from (Fig. 2).

5 PILOT STUDY

To compare with conventional methods, a pilot study with 4 participants (mean age=22.8) is conducted. The participants are asked to play an animal counting game while walking along a designed corridor (Fig. 1). To make the task simple, a red cone is randomly placed in the middle of the road as an obstacle, while three red fire extinguishers are placed at the side of the road as a confusion item. For comparison, we include a Smombie baseline where the participants are asked to use smartphones while walking as usual. Another

Table 1: The study result. D.B.C.:Distance Before Collision, T.B.C.:Time Before Collision. Acc.: Accuracy of detection.

Method	D.B.C.	T.B.C.	Acc.
Smombie	1.08 m	1.71 s	N/A
Direct Detect. [Goh et al. 2020]	5.12 m	8.13 s	85%
Ours (Prediction)	5.30 m	8.48 s	97.5%

direct detection baseline employs the random forest detection algorithm as Goh et al. [Goh et al. 2020]. In the Smombie baseline, the participants are asked to stop when they notice an obstacle, while in the other conditions participants stop when a warning is sent from the phone. The distance before collision (D.B.C) and the user's speed are recorded, the results are shown in Table 1.

For the accuracy test, each participant is asked to pass the route ten times, resulting in 40 tests for each method. From the result, only one miss detection (wrong object detected as obstacle) of 40-time (2.5%) happened in Ours condition, but 6 times (15%) in the direct detection. Overall, our method outperforms the other two methods with a further detection distance and a higher accuracy.

6 DISCUSSION & CONCLUSION

The results from the pilot study suggests that our methods can recognize an obstacle from a mean distance of 5.30m with a high precision, which is 4 meters further than the range of a smombie. However, the current work still faces some limitations. Firstly, the risk map is difficult to generate since it requires manual simulation and an automated script is necessary for general obstacles. Next, the threshold of the risk degree is not studied, experiments are required to decide a proper threshold. Also, a more detailed user study including qualitative and quantitative study of user's preference and real risk reduction need to be performed.

In conclusion, we proposed a novel context-aware risk prediction system which predicts the risk degree of a potential obstacle. We also introduced a synthetic data collection method using a virtual smombie simulator. Different from existing methods which annoy users with frequent notifications, our system use risk degree as a threshold to warn the user only when necessary. The proposed system can reduce smombie accident and also has a potential to be applied to more scenarios such as distracted driving (especially for bikes/motorbikes) or supporting people with vision impairment.

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